

COL100: Introduction to Computer Science

Lab 1: Your First Program

January 6, 2025

1 Setting up Environment

Welcome to your first COL100 Lab! In this lab, you will familiarize yourselves with the tools you will be using to write code throughout the semester. At the end of this lab, you will be familiar with using Anaconda and Gradescope and will have written, tested, and submitted your first Python program!

1.1 Python

Python is a versatile and powerful programming language known for its simplicity and readability, making it an ideal choice for both beginners and professionals. Its extensive libraries and frameworks enable rapid development of applications, from web development to data science and machine learning. Learning Python opens doors to automation, data analysis, and software development, allowing you to solve complex problems efficiently while maintaining clean, easy-to-understand code. Whether you are building a small script or developing a large-scale application, Python is a key tool in the modern programmer's toolkit.

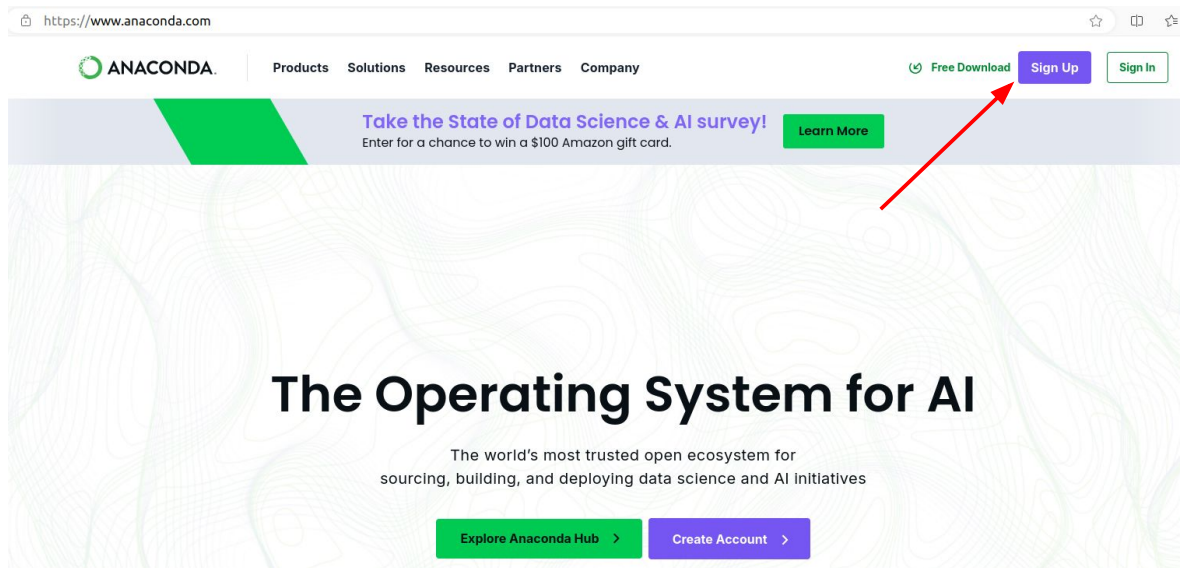
Fun Fact! Some of the world's biggest platforms like Instagram, YouTube, and Dropbox are powered by Python, highlighting its versatility and power in building large-scale applications!

1.2 Anaconda

Anaconda is a beginner-friendly platform specifically designed for Python. It comes with everything you need, including Python, essential libraries, and tools like Jupyter Notebooks, all pre-installed, so you don't have to worry about setup. Anaconda simplifies package management and deployment, making it easier to work with data, machine learning, and scientific computing. This is the tool we will be using in this course to write and run Python code efficiently.

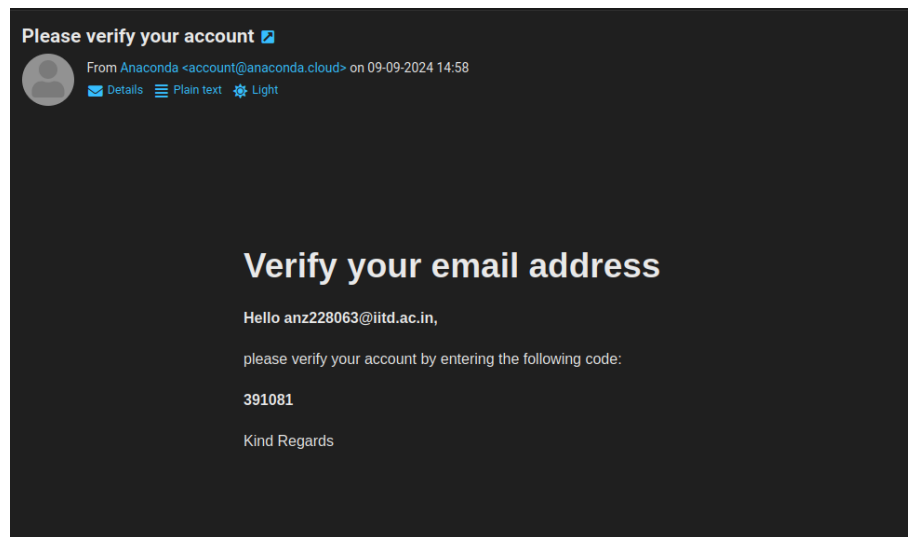
1.2.1 Create an Account on Anaconda

- Visit <https://www.anaconda.com/>
- Click on *Signup*



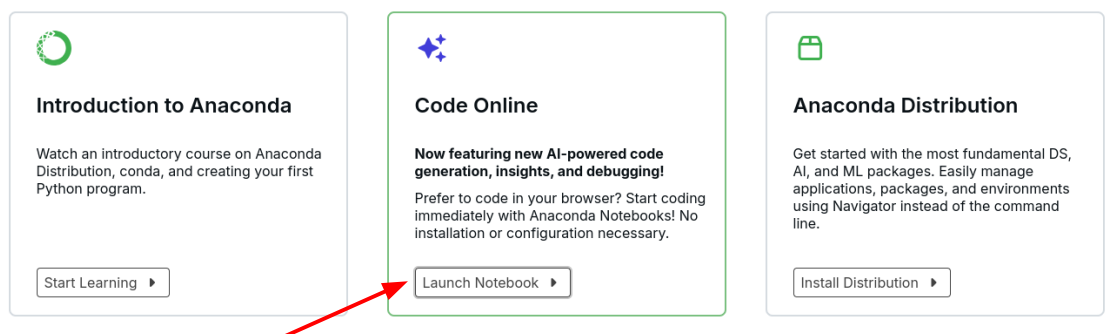
- Click on *Continue with email*

- Enter your *institute email id* and a create a password and then click on create an account.
- You should receive a verification email on your institute email ID.

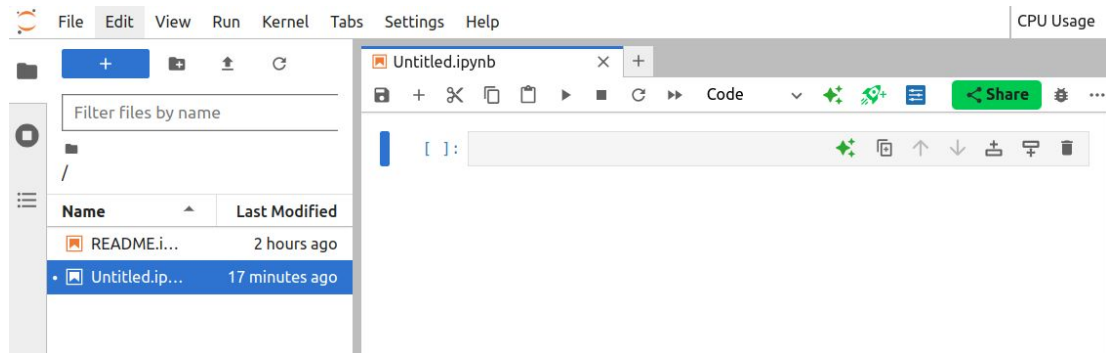


- Click of *Verify Now*. You should be redirected and should be able to see the Anaconda home screen.

Welcome to Anaconda Cloud



- You should then click on “**Launch Notebook**”.



1.3 Basic Linux Terminal Commands

Linux is an open-source operating system that powers a wide range of devices, from servers to smartphones. It is known for its flexibility, security, and extensive support from the global developer community. We will learn about how to interact with Linux command line interface (CLI) on the Anaconda environment in the following sections.

We first need to open the CLI, i.e. terminal, this can be done with the following actions:

- Click on the blue plus button in the top left.
- Under the other option, you will see Terminal. Select that icon.

1.3.1 Basic Commands

- **pwd**: Print Working Directory
Within the interface it is often difficult to remember our location while navigation. We can use **pwd** command to remind us where we are. This is the same as looking at your Finder/Explorer window in a graphics interface.

```
1 pwd
2 Example: /home/ali/Desktop
```

- **ls**: List Directory Contents
When you want to see the files and folders in your current directory, use the **ls** command to list them. This is the same as looking at files and folders in a Finder/Explorer window.

```
1 ls
2 Example: index.html proxy.sh file.txt
```

- **mkdir:** Make Directory

To create a new folder for organizing your files, use the **mkdir** command followed by the desired folder name. This is the same as doing a right mouse click and selecting new folder.

```
1 mkdir my_directory
```

- **cd:** Change Directory

Use the **cd** command to move between folders and navigate the filesystem. This is the same as double clicking on a desired folder to navigate to its contents.

```
1 cd /path/to/directory
```

- **touch:** Create a New File

If you need to quickly create an empty file, the **touch** command is a simple way to do so. The command below will create a new empty file named `myfile.txt`.

```
1 touch myfile.txt
```

- **rm:** Remove File

When you want to delete a file, use the **rm** command with the filename. This is the same as moving something to the recycle bin and permanently deleting it.

```
1 rm myfile.txt
```

1.4 Piazza

Piazza is an interactive, web-based Q&A platform designed to enhance classroom communication and collaboration. It allows students and instructors to post questions, share answers, and discuss topics in real-time, fostering a community-based learning environment. Its organized structure and features like anonymous posting encourage participation and make it easier to manage class discussions effectively.

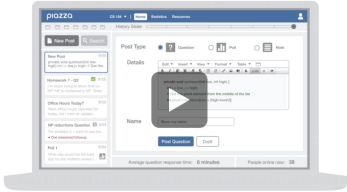
1.4.1 Accessing Piazza

- Visit <https://piazza.com/>

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The incredibly easy, incredibly engaging Q&A platform

Save time and help students learn using the power of community



- Wiki style format enables collaboration in a single space
- Features LaTeX editor, highlighted syntax and code blocking
- Questions and posts needing immediate action are highlighted
- Instructors endorse answers to keep the class on track
- Anonymous posting encourages every student to participate
- Highly customizable online polls
- Integrates with every major LMS

Students Get Started

Professors and TAs Get Started

View a Real Class

[Learn more about how Piazza complies with FERPA](#)

- Click on Sign Up, enter your college name, and select the class you want to join. If you already have an account, click the Log In button and enter your Kerberos email and Piazza account password.

Log In

Email*

Password*

* Required fields

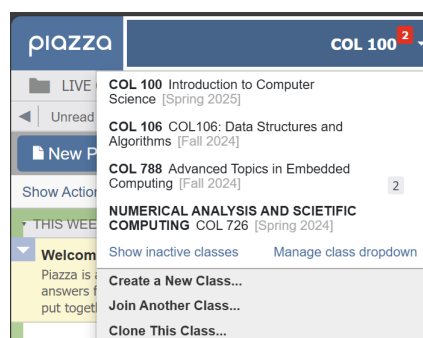
☒ Remember me

Log in

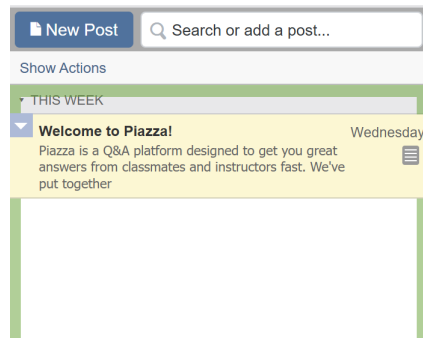
[Forgot your password?](#)

[Login with your school's credentials \(SSO\)](#)

- To join a class, select the Join Other Class option from the dropdown menu where the class names are listed.



- Use the dropdown menu to choose the class you wish to interact with.
- To start a new post, click the New Post button in the top-left corner of your Q&A dashboard. To participate in an existing discussion, click on the relevant post in the bottom section of the dashboard.



- Use the Compose Follow-Up Discussion section at the bottom of any post to continue the conversation. You can format your message and choose whether to post it anonymously. Click on submit after formatting.

1.4.2 Submission instructions

- Save the screenshot of your piazza post.
 1. Press Fn + Alt + PrtScr to save the active screen as a picture on your terminal
 2. Go to Pictures directory in your home directory

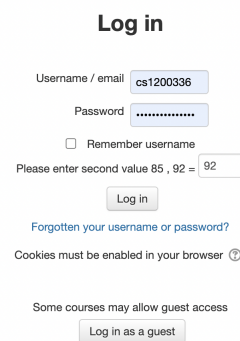
3. The most recent picture should be the one you need

- Copy the screenshot in a document i.e. Insert the picture mentioned in the previous step.

1.5 Moodle

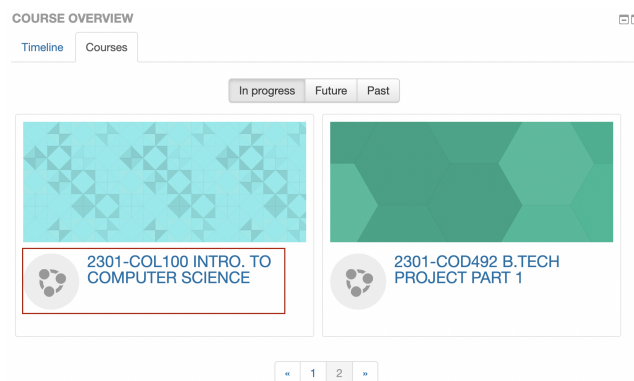
1.5.1 Logging into Moodle

- Go to <https://moodle.iitd.ac.in/>
- Use your Kerberos ID and password to log in.

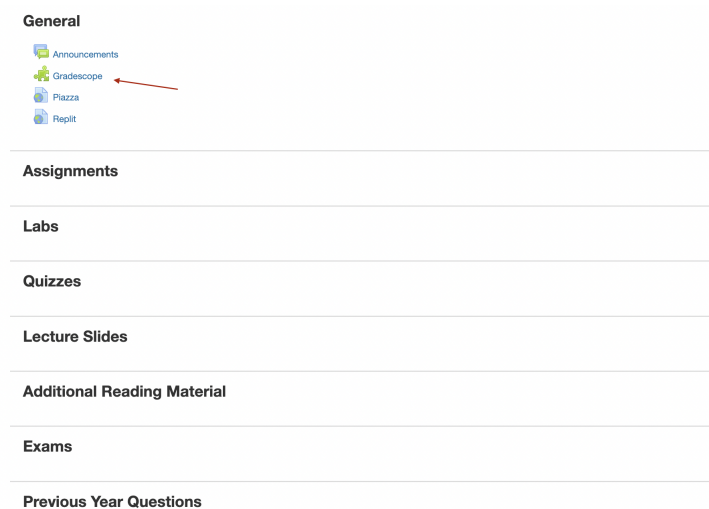


The login form is titled "Log in". It contains a "Username / email" field with the value "cs1200336", a "Password" field with masked characters, and a "Remember username" checkbox. Below these is a CAPTCHA prompt: "Please enter second value 85 , 92 =" followed by a text box containing "92". A "Log in" button is positioned below the CAPTCHA. There is a link for "Forgotten your username or password?". At the bottom, it says "Cookies must be enabled in your browser" with a help icon, and "Some courses may allow guest access" with a "Log in as a guest" button.

- After logging in, your courses will be listed under the My Courses section on the Dashboard. Select the desired course by clicking on its name.



- On the course homepage, you'll find useful links at the top of the page. Scroll down to access lecture notes, assignment links, and other resources organized by date.



1.6 Gradescope

Next, we'll get familiar with the flow you need to follow for submitting assignments.

- You should have added to Gradescope.
- Go to the page for “COL100: Introduction to Computer Science” and join the class.
- Go to the “Assignments” tab and select the appropriate assignment. In this case, you can select Lab 1.
- Recall that you can write code, run it, and test it using Anaconda. For the final submission, you need to download your code from Anaconda and submit it on Gradescope.
- Your code will be automatically evaluated against test cases on Gradescope.

2 Your First Lab

2.1 Let's Try Some Linux:

1. Using the anaconda.cloud open a terminal window.
2. Create a directory named <YourEntryNumber>_COL100 (Eg: 2024CS10001_COL100).
3. Inside <YourEntryNumber>_COL100, create a file named `hello.txt`.
4. Rename the file to <YourEntryNumber>_Lab1.py.
Example: 2024CS10001_Lab1.py

2.2 Let's write a Program

1. Using the web interface, navigate to the file you created.
2. Write the following code in <YourEntryNumber>_Lab1.py:

```
1 print("Hello, world!")
```
3. Save your program.
4. Navigate back to the terminal and traverse to the file location.
5. Execute your Python file:

```
1 python <YourEntryNumber>_Lab1.py
```
6. Verify that the output shows: `Hello, world!`

Congratulations! You have successfully written and run your first program. Feel free to practice and learn programming using this flow.

2.3 Submission Instructions

1. For this lab, you need to submit the following three files on Gradescope.
 - (a) Python Program
 - Use your original <YourEntryNumber>_Lab1.py file
 - Ensure it contains the “Hello, world!” program
 - The program should run without any errors
 - (b) Moodle Screenshot for Access Verification:

- Access the COL100 Moodle page
- Take a screenshot (Fn + Alt + PrtScr)
- Save as <YourEntryNumber>_moodle.png

(c) Piazza Screenshot for Access Verification:

- Make a post introducing yourself on Piazza
- Take a screenshot of your post (Fn + Alt + PrtScr)
- Save as <YourEntryNumber>_piazza.png

2. Submission Steps:

(a) Go to Gradescope's "Lab 1" assignment

(b) Upload all three files:

- <YourEntryNumber>_Lab1.py
- <YourEntryNumber>_moodle.png
- <YourEntryNumber>_piazza.png

(c) Submit and verify all files are uploaded correctly

3 Practice Problems

3.1 Mathematical Operations in Python

Write a Python program that calculates and displays the result of the following mathematical expression: $2^{15} - 10000 * 2$

3.1.1 Instructions:

1. Create a new Python file named `<YourEntryNumber>_practice.py`
2. Write code to calculate the above expression
3. Use the `print()` function to display the result
4. Your output should show both the calculation and its result

3.1.2 Example Output Format:

```
1 The result of 2^15 - 10000 * 2 is: -19999.8666
```